Nitesh Reddy Thota

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PROFESSIONAL SUMMARY

AI Engineer with 3+ years building production data agents and LLM pipelines that answer real questions from real data. Built **HAWK AI** - a Claude-powered knowledge agent over ServiceNow incident history with semantic chunking, FAISS vector search, and <500ms p95 inference - and a GPT-4 agentic triage system processing 500+ tickets/week at 90%+ accuracy. Own the full stack: data ingestion, RAG architecture, LLM orchestration (tool/function calling, prompt versioning), eval telemetry (RAGAS), and Grafana observability dashboards. Rapid prototyper and production engineer in equal measure. STEM OPT authorized through 2027.

CORE COMPETENCIES

AI Agents & Agentic Architecture | Production RAG Pipelines | LLM Orchestration & Tool Calling | Data Agent Design | LLM Architecture Decisions | LLMOps & Eval Telemetry (RAGAS) | Backend Engineering (AWS Lambda/EKS) | Cloud Data Platforms (Kinesis/Glue/Athena) | Prototype-to-Production | HIPAA-Compliant AI Systems

WORK EXPERIENCE

JunoLabs / Resolve Tech Solutions (Client: Healthcare) — Jan 2026 - Present

AI/ML Engineer Intern · Richardson, TX

- Built **HAWK AI** - a production **data agent** over ServiceNow incident history powered by **Claude via Amazon Bedrock**: document ingestion, semantic chunking, text embeddings, and FAISS vector search enabling automated resolution retrieval at **<500ms p95 inference latency**.
- Engineered prompt/context patterns and **tool/function calling** workflows including structured output schemas, multi-turn context management, and prompt versioning registry - reducing prompt regression incidents ~60% and cutting token cost per inference ~35%.
- Operationalized **LLMOps governance**: Bedrock ARN pinning per model version, RAGAS-style evaluation telemetry (relevance + faithfulness), A/B prompt testing framework, and Grafana dashboards tracking retrieval quality across every release.
- Implemented trust, safety, and compliance controls: PII detection/redaction pre-prompt, prompt-injection defenses, content filtering (HIPAA/SOC 2-aligned), and audit logging of all LLM interactions.
- Architected end-to-end AWS **analytics platform** - Kinesis → Lambda → S3 → Glue → Athena - ingesting Amazon Connect CTR data from 3+ live sources, serving a real-time help-desk dashboard for 4 role-based tiers; reduced load latency 65% via CloudFront + automated cache invalidation.
- Unified 5 analytics actions behind a single Lambda + API Gateway endpoint with Athena partition projection - slashing query latency 70% and data retrieval costs 40%.
- Cut CI/CD cycle time 55% via branch/path-triggered GitLab pipeline with Docker + zero-downtime EKS Helm rollouts across DEV, UAT, and PROD.
- Implemented AWS Cognito JWT RBAC for 4 user tiers; resolved a critical NAT gateway outage within 2 hrs, preventing 8+ hrs of downstream data loss.

George Mason University ITS — Nov 2024 - Jan 2026

Student Data Engineer · Fairfax, VA

- Built a production **LLM triage agent** using **GPT-4 + LangChain** with function calling for structured ticket classification - processing 500+ tickets/week at **90%+ accuracy**, eliminating 85% of manual categorization.
- Architected 4 Python microservices automating TeamDynamix REST ingestion → parameterized Airflow DAGs → Redshift/PostgreSQL - saving ~8 hrs/week of analyst effort and accelerating reporting 3×.
- Hardened pipeline security via HashiCorp Vault dynamic secrets (zero credential-related incidents); elevated observability with Prometheus + Grafana across 6 services, cutting MTTD 50%.
- Cut Docker + EKS Helm-chart deployment time 60% for the GPT-4 classification pipeline.

Modak Analytics (Client: Humana) — Jul 2021 - Jul 2023

Data Engineer · Hyderabad, India

- Scaled HIPAA healthcare claims processing to **10+ TB/day** via distributed PySpark + Apache Beam on GCP Dataflow/Dataproc; reduced batch runtime 35% via windowed aggregation optimization.
- Cut BigQuery costs 45% (~$18K/year) and reduced p95 latency from 12s → 2s via partition pruning, clustering, and materialized views; achieved 99.8% data quality SLA (Great Expectations + Pub/Sub - 1,200+ anomalies/month intercepted).
- Co-designed hydration framework accelerating data onboarding 4× across 5 heterogeneous sources (Oracle, PostgreSQL, flat files).
- Eliminated ~15% of downstream failures via automated Avro/Parquet schema evolution with GCS + Hive Metastore.

PROJECTS

PatriotPilot - RAG & Document Intelligence Agent Production-grade

Production document intelligence system with semantic chunking (512-token overlapping windows), E5-Large-v2 embeddings, FAISS vector search, and hybrid search (dense + BM25 sparse) for sub-100ms retrieval. RAGAS evaluation telemetry (relevance, faithfulness, context precision) on 500+ queries. Fine-tuned Qwen 2.5 14B with LoRA + DeepSpeed ZeRO - 70%+ Top-1 accuracy on benchmark.

Python · LlamaIndex · FAISS · RAGAS · LoRA · DeepSpeed · HuggingFace

UAV Decision Guidance System - Agentic Trajectory Optimization

Multi-objective path planning framework using binary decision variables and 3D leg metrics. Structured tool-calling patterns for autonomous decision execution. KPI-based routing with patrol mode visualization; integrated with PX4-ROS2-Gazebo simulation.

Python · ROS2 · PX4 · Gazebo · structured tool calling

EDUCATION

M.S. Computer Science - George Mason University, Fairfax, VA — Aug 2023 - May 2025

GPA: 3.87/4.0 · Coursework: Data Mining, NLP, Database Systems, Software Engineering, Decision Guidance Systems (A+)

B.Tech. Electronics & Communication Engineering - SRM Institute, India — Jul 2017 - Jun 2021

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (Sep 2025)

AWS Certified Machine Learning - Specialty (Dec 2025)

TECHNICAL SKILLS

GenAI & LLM: Claude (Anthropic/Bedrock), GPT-4, Gemini; LangChain, LlamaIndex; RAG, Vector/Hybrid Search, Semantic Chunking, Tool/Function Calling, Prompt Engineering, RAGAS, LLMOps, Prompt-Injection Defenses, PII Redaction

Cloud: AWS (Bedrock, SageMaker, EKS, Kinesis, Glue, Athena, S3, CloudFront, Cognito, Lambda, API Gateway); GCP (Dataflow, Dataproc, BigQuery, Vertex AI); Azure (ADF)

Data Engineering: Apache Spark, Kafka, Airflow, Beam; PostgreSQL, Redshift, BigQuery, Athena, Snowflake; FAISS, Pinecone; Avro, Parquet, Hive Metastore

Backend & DevOps: Python, SQL, REST, GraphQL, Microservices; Docker, Kubernetes, Helm, GitLab CI/CD, Terraform; Prometheus, Grafana; HashiCorp Vault

ML & Analytics: Scikit-learn, TensorFlow, PyTorch, BigQuery ML, LoRA fine-tuning, DeepSpeed; Power BI, Tableau